

WEEKLY PROGRESS REPORT – WEEK 5

Project Title

Healthcare Analytics Pipeline for MediCare Hospital

Group Number: 3

Team Members

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Week 5 Objectives

The objective of Week 5 was to integrate all project components, enhance dashboard functionality, validate cloud deployment, update project documentation, and prepare the project for final submission.

Activities Completed

- 1. Project Integration**
All project modules including ETL pipelines, cloud services, data warehouse components, machine learning workflows, and dashboards were integrated into a single end-to-end healthcare analytics solution.
- 2. Dashboard Enhancement and Testing**
The Streamlit dashboard was enhanced with interactive filters, KPI visualizations, and analytical charts. Dashboard functionality and responsiveness were tested across different scenarios.
- 3. Cloud Deployment Verification**
The cloud-based dashboard deployment was validated to ensure accessibility, reliability, and proper functioning of analytical features.
- 4. GitHub Repository Organization**
The project repository was updated with documentation, screenshots, reports, and supporting project resources to maintain proper project structure and version control.
- 5. Documentation Updates**
Project documentation including README updates, dashboard screenshots, architecture diagrams, and implementation details were reviewed and finalized.

6. System Testing and Validation

End-to-end testing was conducted to validate ETL execution, dashboard analytics, database integrations, and machine learning outputs.

7. Team Coordination and Review

The team conducted review meetings to verify deliverables, discuss pending tasks, and prepare for final submission activities.

Deliverables Completed

- Integrated healthcare analytics pipeline
- Enhanced Streamlit dashboard
- Dashboard screenshots and analytics views
- Updated GitHub repository
- Updated project documentation
- End-to-end testing and validation report
- Deployment verification activities

Challenges Faced

- Integrating multiple technologies into a single workflow
- Validating dashboard deployment and accessibility
- Maintaining documentation consistency
- Verifying end-to-end system functionality
- Coordinating final project activities

Plan for Week 6

- Finalize project report and presentation
- Record project demonstration video
- Complete final repository updates
- Submit all project deliverables
- Conduct final team review and project closure

Conclusion

Week 5 focused on project integration, dashboard enhancement, deployment verification, and documentation updates. The team successfully integrated all project components and prepared the Healthcare Analytics Pipeline for final submission and project closure.